MATH210B Homework 9

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1 Problem 1

Let $\alpha \in K \backslash F$ and consider the minimal polynomial p of α over F.

Since $\alpha \notin F$, p is not separable. Then, $p' = 0$. Then, $p(x) = f(x^p)$ for some $x \in F[x]$.

Notice that f also needs to irreducible, so if f is not separable, we can keep applying this argument to f.

Since K/F is a finite extension, there's some $n > 0$ such that $p(x) = f(x^{p^n})$ and f is separable and irreducible. Then, f has to split since the extension is purely inseparable. But then, f has a single linear factor since otherwise f would be reducible.

Then, $f(x) = x^{p^n} - a$ for some $a \in F$. But then $\alpha^{p^n} = a$ which is what we wanted to show.

2 Problem 2

Let $\alpha \in E^G$ and assume that α is separable over F.

Then, $\alpha \in K$ is also separable over F.

Let m_α be the minimal polynomial of α.

Our goal is to show that m_α is $x - \alpha$.

m_α splits over E since it's a normal extension and it already contains one root, namely, α. Assume by contradiction that it has another root β that is distinct from α.

Then, we have a field isomorphism $F(\alpha) \to F(\beta)$ by Homework 7 Problem 6.

3 Problem 3

Lemma 3.1. Let K/F be a finite field extension and L/F be its maximal separable subextension. Then, K/L is purely inseparable.

Proof. Let $\alpha \in K$ be separable over L. Then, m_α is separable over L, so it's separable over F. Then, $\alpha \in L$ by definition. $\square$

4 Problem 4

Lemma 4.1. Let K/F be an algebraic field extension. Then, every field homomorphism $\sigma : K \to K$ over F is an isomorphism.

Proof. σ is injective since σ fixes F and fields have no non-trivial ideals.

Let $\alpha \in K$.

Let $m_\alpha \in F[x]$ be the minimal polynomial of α.

Let L be the subfield of K generated by the roots of m_α that is already in K. Notice that L/F is a finite extension, and thus the restriction of σ to L is an isomorphism using Rank-Nullity, so σ maps onto α. $\square$

5 Problem 5

Lemma 5.1.

6 Problem 6

Lemma 6.1. F is a perfect field if and only if every irreducible polynomial f over F is separable.

Proof. This result is trivial if $char(F) = 0$. Thus, assume $char(F) = p > 0$.

Assume F is perfect. Let $f \in F[x]$ be irreducible.

Assume by contradiction that f is not separable. Then, there's some $g(x) \in F[x]$ such that $f(x) = g(x^p)$ such that $g(x)$ is irreducible.

Let $g(x) = a_n x^n + \cdots + a_1 x^1 + a_0$. Then, $g(x^p) = a_n x^{np} + \cdots + a_1 x^p + a_0$. Since F is perfect, for every $i \in \{1, \ldots, n\}$, we have some $b_i \in F$ such that $b_i^p = a_i$.

Then, $g(x^p) = b_n^p x^{np} + \cdots + b_1^p x^p + b_0^p = (b_n x^n + \cdots + b_1 x + b_0)^p$, which contradicts the irreducibility of F.

Let's now prove the backwards direction. We'll use the contrapositive. Assume F is not perfect. Then, there's some $x \in F$ such that it doesn't have a pth root. Then, $f(t) = t^p - x$ is irreducible over $F[x]$ since we can't factor it as $(t - x)^p$, but this is the factorization in the splitting field which is unique. But then, f is not separable and irreducible. $\square$

The lemma above implies that every algebraic extension of a perfect field is perfect since every irreducible polynomial over in $K[x]$ is separable over K if and only if it is separable over F.

7 Problem 7

Let K/F be a finite field extension.

Part (a) follows immediately by the additivity of trace for matrices and multiplicativity of the determinant.

8 Problem 9